College Internship & Placement Management System

DBMS Lab – Project Based Learning (PBL)

Project Type: Real-life Database Management System
Technology: MySQL 8.x

Prepared by: _____
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1. Problem Statement

Colleges receive internship and placement opportunities from multiple companies. Students need to track companies, job roles, eligibility criteria, skills, applications, and selection status. Managing this information manually can cause duplication, difficulty in tracking applications, and delays in identifying eligible students. The proposed system stores and manages this information in a relational database.

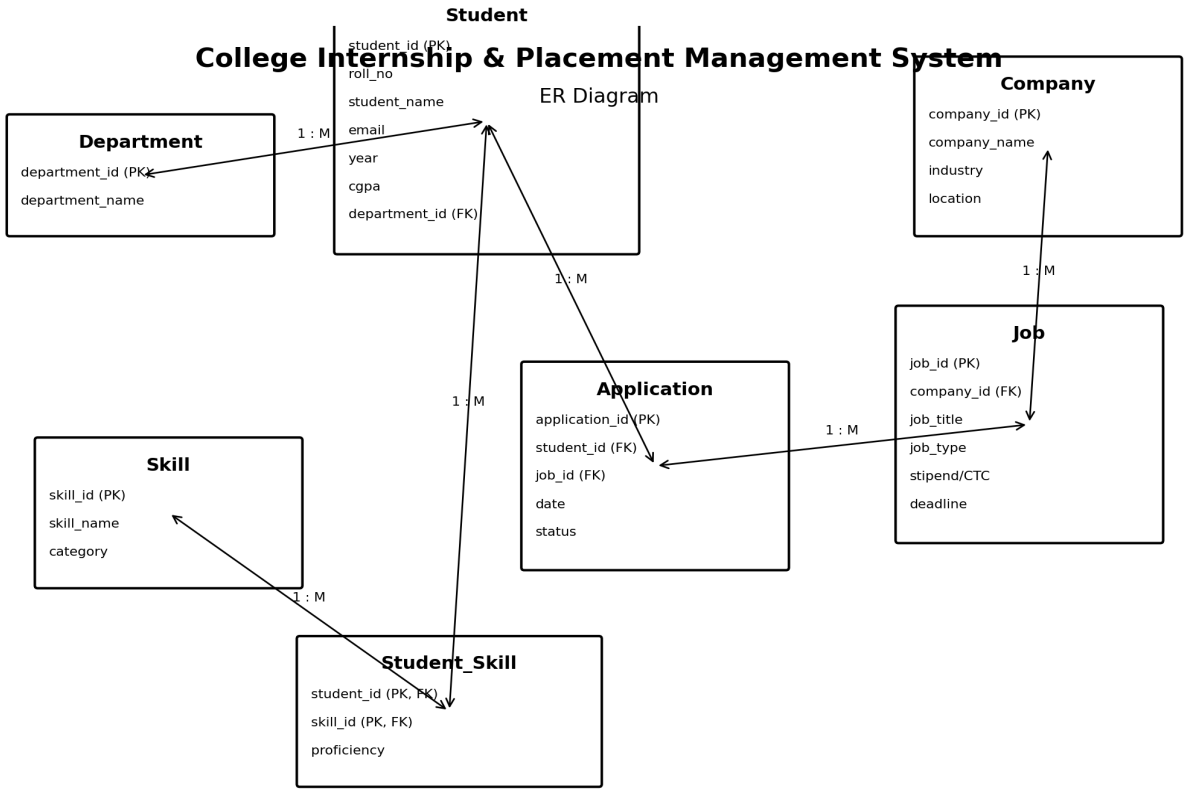
2. Objectives

• Store student and department information.
• Maintain company and job/internship details.
• Record student skills and proficiency levels.
• Track student applications and their status.
• Use SQL joins, aggregation and subqueries to generate useful reports.

3. Scope

The system is intended for a college placement/internship cell. It can later be extended with login, resume storage, interview scheduling, notifications and a web/mobile interface.

4. ER Diagram



Main relationships: one department has many students; one company has many jobs; students can apply to many jobs through Application; students can have many skills through Student_Skill.

5. Relational Schema

Table	Primary Key	Important Foreign Keys
Department	department_id	—
Student	student_id	department_id → Department
Company	company_id	—
Job	job_id	company_id → Company
Skill	skill_id	—
Student_Skill	student_id + skill_id	student_id → Student; skill_id → Skill
Application	application_id	student_id → Student; job_id → Job

6. Normalization

The database separates departments, students, companies, jobs and skills into related tables. Many-to-many student/skill relationships are resolved using Student_Skill, while student/job applications are represented using Application. This reduces redundancy and supports update consistency.

7. SQL Implementation

The accompanying SQL file creates the database, tables, constraints and sample records. It also contains demonstration queries using JOIN, LEFT JOIN, GROUP BY, subquery, VIEW and UPDATE.

Core tables:

• Department

• Student

• Company

• Job

• Skill

• Student_Skill

• Application

8. Sample SQL Queries

Student + Department

```
SELECT s.roll_no, s.student_name, d.department_name, s.cgpa
FROM Student s JOIN Department d ON s.department_id=d.department_id;
```

Applications + Company + Job

```
SELECT s.student_name, c.company_name, j.job_title, a.status
FROM Application a JOIN Student s ON a.student_id=s.student_id
JOIN Job j ON a.job_id=j.job_id JOIN Company c ON j.company_id=c.company_id;
```

Selected Students

```
SELECT s.student_name, c.company_name, j.job_title
FROM Application a JOIN Student s ON a.student_id=s.student_id
JOIN Job j ON a.job_id=j.job_id JOIN Company c ON j.company_id=c.company_id
WHERE a.status='Selected';
```

Applications per Job

```
SELECT j.job_title, COUNT(a.application_id) AS total_applications
FROM Job j LEFT JOIN Application a ON j.job_id=a.job_id
GROUP BY j.job_id, j.job_title;
```

9. Expected Outputs

When the SQL file is executed in MySQL Workbench/phpMyAdmin, the student can demonstrate the following outputs: student lists with departments, job listings with companies, eligible applicants, complete application tracking, application counts, selected students, skill-based searches and average CGPA by department.

10. Conclusion

The College Internship & Placement Management System demonstrates how a relational database can solve a practical college-management problem. The project uses primary keys, foreign keys, constraints, many-to-many relationships, joins, aggregate functions, subqueries and a view. The design can be extended into a full web application in future.

11. Future Enhancements

- Student login and profile management
- Resume upload and parsing
- Automated eligibility matching
- Interview scheduling
- Email/SMS notifications
- Admin dashboard and analytics

Note: Replace the blank student details on the cover page before submission.